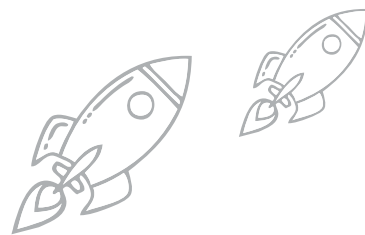


Getting Started with Kubernetes: The Ultimate Guide



Learn how to get started with Kubernetes with the help of this simple tutorial.



An orchestrator is a piece of software that distributes container workloads over several servers or nodes. Its primary function is to orchestrate container workloads. To prevent any kind of downtime from occurring, it is necessary to ensure that the containers have access to all the appropriate resources, that the correct number

of containers is always on hand, and that additional steps, such as rolling deployments, are done. Kubernetes is the container orchestrator that has had the highest amount of adoption recently.

Before moving on to Kubernetes, let's get a grasp on containers and see how they work. Containerization refers to the process of containing the source code of an application as well as any

dependencies it has into a single image, and then running that image within a computer environment that is isolated from any other environments.

It is common practice to credit Docker as the breakthrough that led to the development of containerization. This is because Docker made it possible to package software into a portable virtual container. Containerization provides the portability of code to cloud services such as Google Cloud Platform, Amazon Web Services, and Digital Ocean.

However, complications arise as the number of containers increases. Deploying and managing hundreds of containers across several servers or virtual machine instances can be a very time-consuming process. That's what led to the development of container orchestration tools like Kubernetes.

Kubernetes, which is often referred to as K8s, was first released by Google in 2015, and is now being maintained by an open source community (of which Google is also a member). Kubernetes is a system that, by using a combination of application programming interfaces and command line tools, enables the coordination of container processes that span several hosts. The process of deploying, scaling, and maintaining container clusters across several hosts is simplified as a result of using this tool. Because it handles problems like configuration management,